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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning of Café Sales Dataset

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Data Cleaning using Power Query

GitHub Link: <https://github.com/keerthi19hub/Business-Analytics>

Introduction

The Cafe Sales dataset is a raw dataset containing café sales information such as transactions, items sold, quantities, prices, payment methods, and other sales details. The dataset was obtained from Kaggle and was intentionally provided in a “dirty” and uncleaned form to simulate real-world business data.

The dataset contained various data quality issues, including missing values, inconsistent text entries, incorrect data types, calculation errors, and duplicate records. These issues could affect the accuracy of analysis if not properly addressed.

To resolve these issues, Microsoft Power Query was used to clean and transform the dataset. Different transformations were applied and recorded as Applied Steps, making the process organized and easy to track. The main objective was to convert the raw Cafe Sales dataset into a clean, consistent, and analysis-ready dataset.

2. Dataset Overview

The raw dataset consisted of eight columns and close to one thousand rows of transactional cafe sales records. The key fields in the dataset are Item (the product sold), Quantity (number of units sold), Price Per Unit (unit price of the item), Total Spent (total transaction value, equal to Quantity multiplied by Price Per Unit), Payment Method (Cash, Credit Card or Digital Wallet), Location (In-store or Takeaway), and Transaction Date (the date on which the sale occurred). After cleaning, Quantity was stored as a whole number, Price Per Unit and Total Spent as decimal numbers, Transaction Date as a proper date, and the remaining fields as text.

The following issues were identified in the raw data before cleaning began:

- Missing (blank) values across several columns, including Item, Payment Method, Location, Quantity and Price Per Unit.
- Placeholder text such as “UNKNOWN” and “ERROR” used in place of genuine missing values.
- Incorrect or inconsistent data types, with numbers and dates stored as plain text.
- Mathematical inconsistencies between Quantity, Price Per Unit and Total Spent.

- Duplicate transaction records.
- Extra spaces and inconsistent casing in text fields.
- Invalid or out-of-range transaction dates and negative or zero quantities.

3. Tools and Approach

The cleaning process was carried out entirely in the **Power Query Editor**, available through Excel's "**Get & Transform Data**" or Power BI's "**Transform Data**" option. Power Query was used because it records each transformation as an **Applied Step**, making the process transparent, organized, and repeatable. It also preserves the original source data and allows the cleaning steps to be automatically reapplied when the data is refreshed.

The cleaning process followed four main phases: **import and data type correction, error and placeholder handling, column-level corrections, and row-level filtering and deduplication**. Each phase and its individual steps are explained in the following sections.

4. Step-by-Step Data Cleaning Process

A total of twenty-three Applied Steps were used to clean the dataset, as shown in the Power Query Query Settings pane. The steps, in sequence, are: Initial Type, Headers, Set Data Types, Replace Error, Replace UNKNOWN, Format Types, Fix Total Spent, Fix Quantity, Fix Price, Remove Originals, Rename Columns, Final Data Types, Filter Dates, Fill Item, Fill Payment, Fill Location, Remove Invalid Rows, Remove Duplicates, Trim Text, Clean Text, Remove Blanks, Filter Quantity, and Final Filter.

4.1 Detailed Explanation of Each Step

Step 1: Loading the Raw Data

Power Query Applied Step: Initial Type

The dataset was imported into Power Query directly from the source CSV file. At this point every column was automatically read as Text, which is Power Query's default behaviour for freshly imported data. This step establishes the untouched starting point of the query before any cleaning logic is applied.

Table.TransformColumnTypes(Source, [{"Column1", type text}, {"Column2", type text}, {"Column3", type text}, {"Column4", type text}, {"Column5", type text}, {"Column6", type text}, {"Column7", type text}], ...)

Column1	Column2	Column3	Column4	Column5	Column6	Column7
Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card	In-store
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway
TXN_6695534	Sandwich	4	4.0	16.0	Cash	UNKNOWN
TXN_4717867		5	3.0	15.0		Takeaway
TXN_2064365	Sandwich	5	4.0	20.0		In-store
TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway
TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway
TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store
TXN_9437049	Cookie	5	1.0	5.0		Takeaway
TXN_8915701	ERROR	2	1.5	3.0		In-store
TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store
TXN_3765707	Sandwich	1	4.0	4.0		
TXN_6769710	Juice	2	3.0	6.0	Cash	In-store
TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR
TXN_3709394	Juice	4	3.0	12.0	Cash	Takeaway

Query Settings: Name: dirty_cafe_sales

APPLIED STEPS: Source, Initial Type, Headers, Set Data Types, Replace Error, Replace UNKNOWN, Format Types, Fix Total Spent, Fix Quantity, Fix Price, Remove Originals, Rename Columns, Final Data Types, Filter Dates, Fill Item, Fill Payment, Fill Location, Remove Invalid Rows, Remove Duplicates, Trim Text, Clean Text, Remove Blanks, Filter Quantity

Step 2: Promoting the First Row to Headers

Power Query Applied Step: Headers

The first row of the raw file, containing column names such as Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location and Transaction Date, was promoted to become the table's column headers using "Use First Row as Headers." This ensured that all subsequent steps could reference columns by their proper names rather than generic "Column1, Column2" labels.

Table.PromoteHeaders("#Initial Type", [PromoteAllScalars=true])

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card	In-store
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway
TXN_6695534	Sandwich	4	4.0	16.0	Cash	UNKNOWN
TXN_4717867		5	3.0	15.0		Takeaway
TXN_2064365	Sandwich	5	4.0	20.0		In-store
TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway
TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway
TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store
TXN_9437049	Cookie	5	1.0	5.0		Takeaway
TXN_8915701	ERROR	2	1.5	3.0		In-store
TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store
TXN_3765707	Sandwich	1	4.0	4.0		
TXN_6769710	Juice	2	3.0	6.0	Cash	In-store
TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR
TXN_3709394	Juice	4	3.0	12.0	Cash	Takeaway

Query Settings: Name: dirty_cafe_sales

APPLIED STEPS: Source, Initial Type, Headers, Set Data Types, Replace Error, Replace UNKNOWN, Format Types, Fix Total Spent, Fix Quantity, Fix Price, Remove Originals, Rename Columns, Final Data Types, Filter Dates, Fill Item, Fill Payment, Fill Location, Remove Invalid Rows, Remove Duplicates, Trim Text, Clean Text, Remove Blanks, Filter Quantity

Step 3: Assigning Initial Data Types

Power Query Applied Step: Set Data Types

Each column was assigned an appropriate data type: Whole Number for Quantity, Decimal Number for Price Per Unit and Total Spent, Date for Transaction Date, and Text for Item, Payment Method and Location. Because several cells contained non-numeric placeholder text such as "ERROR" or "UNKNOWN," this conversion produced a number of Error values, which were handled in the next step.

action ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
	100% Valid 0% Error 0% Empty	97% Valid 0% Error 3% Empty	99% Valid 0% Error 1% Empty	98% Valid 0% Error 2% Empty	98% Valid 0% Error 2% Empty	74% Valid 0% Error 24% Empty
10 distinct, 10000 unique	11 distinct, 0 unique	8 distinct, 0 unique	9 distinct, 0 unique	20 distinct, 0 unique	6 distinct, 0 unique	5 distinct, 0 unique
373	Coffee	2	2.0	4.0	Credit Card	Takeaway
1031	Cake	4	3.0	12.0	Cash	In-store
903	Cookie	4	1.0	ERROR	Credit Card	In-store
1554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN
1411	Coffee	2	2.0	4.0	Digital Wallet	In-store
1893	Smoothie	5	4.0	20.0	Credit Card	
1211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway
1934	Sandwich	4	4.0	16.0	Cash	UNKNOWN
1867		5	3.0	15.0		Takeaway
1365	Sandwich	5	4.0	20.0		In-store
1360	Salad	5	5.0	25.0	Cash	Takeaway
279	Sandwich	2	4.0	8.0	Credit Card	Takeaway
1095	Sandwich	2	4.0	8.0	Cash	In-store
1049	Cookie	5	1.0	5.0		Takeaway
1701	ERROR	2	1.5	3.0		In-store
1255	Salad	3	5.0	15.0	Credit Card	In-store
1707	Sandwich	1	4.0	4.0		
1710	Juice	2	3.0	6.0	Cash	In-store
1618	Cake	5	3.0	15.0	Cash	ERROR
1094	Juice	4	3.0	12.0	Cash	Takeaway

Step 4: Converting Errors into Nulls

Power Query Applied Step: *Replace Error*

Cells that could not be converted to their target data type, for example text such as “ERROR” inside a numeric column, were flagged by Power Query as Error values. Using “Replace Errors,” all such error values were converted into standard null (blank) values. This is a critical step because Power Query cannot evaluate or filter a column that still contains unresolved errors.

Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction ID
94% Valid 0% Error 6% Empty	97% Valid 0% Error 3% Empty	96% Valid 0% Error 4% Empty	97% Valid 0% Error 3% Empty	71% Valid 0% Error 28% Empty	64% Valid 0% Error 34% Empty	
10 distinct, 0 unique	7 distinct, 0 unique	8 distinct, 0 unique	19 distinct, 0 unique	5 distinct, 0 unique	4 distinct, 0 unique	367 distinct, 0 unique
1 Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
2 Cake	4	3.0	12.0	Cash	In-store	2023-05-16
3 Cookie	4	1.0		Credit Card	In-store	2023-07-19
4 Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
5 Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
6 Smoothie	5	4.0	20.0	Credit Card		2023-03-31
7 UNKNOWN	3	3.0	9.0		Takeaway	2023-10-06
8 sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
9	5	3.0	15.0		Takeaway	2023-07-28
10 sandwich	5	4.0	20.0		In-store	2023-12-01
11 salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
12 sandwich	2	4.0	8.0	Credit Card	Takeaway	
13 sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
14 Cookie	5	1.0	5.0		Takeaway	2023-06-01
15	2	1.5	3.0		In-store	2023-03-21
16 salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
17 sandwich	1	4.0	4.0			2023-06-10
18 juice	2	3.0	6.0	Cash	In-store	2023-02-04
19 cake	5	3.0	15.0	Cash		2023-03-25
20 juice	4	3.0	12.0	Cash	Takeaway	2023-01-15

Step 5: Standardizing Placeholder Text

Power Query Applied Step: *Replace UNKNOWN*

The dataset used the literal string “UNKNOWN” in several text columns, including Item, Payment Method and Location, to represent missing information. These values do not behave like true blanks and would otherwise be treated as valid data. The “Replace Values” transformation was used to convert every occurrence of “UNKNOWN” into a proper null, so that missing data could be consistently identified and handled later using fill or filter operations.

Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
1 Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
2 Cake	4	3.0	12.0	Cash	In-store	2023-05-16
3 Cookie	4	1.0		Credit Card	In-store	2023-07-19
4 Salad	2	5.0	10.0			2023-04-27
5 Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
6 Smoothie	5	4.0	20.0	Credit Card		2023-03-31
7	3	3.0	9.0		Takeaway	2023-10-06
8 Sandwich	4	4.0	16.0	Cash		2023-10-28
9	5	3.0	15.0		Takeaway	2023-07-28
10 Sandwich	5	4.0	20.0		In-store	2023-12-31
11 Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
12 Sandwich	2	4.0	8.0	Credit Card	Takeaway	
13 Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
14 Cookie	5	1.0	5.0		Takeaway	2023-06-01
15	2	1.5	3.0		In-store	2023-03-21
16 Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
17 Sandwich	1	4.0	4.0			2023-06-10
18 Juice	2	3.0	6.0	Cash	In-store	2023-02-24
19 Cake	5	3.0	15.0	Cash		2023-03-25
20 Juice	4	3.0	12.0	Cash	Takeaway	2023-01-15
21 Smoothie		4.0	20.0	Cash	In-store	2023-04-04

Step 6: Standardizing Formats

Power Query Applied Step: *Format Types*

With errors and placeholders resolved, all numeric and date columns were reformatted to a consistent standard, such as a fixed date format and two-decimal formatting for currency values. This step ensures uniform formatting across the entire table irrespective of how each value was originally entered in the source file.

Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
1 Coffee	2	2	4	Credit Card	Takeaway	
2 Cake	4	3	12	Cash	In-store	
3 Cookie	4	1		null Credit Card	In-store	
4 Salad	2	5	10			
5 Coffee	2	2	4	Digital Wallet	In-store	
6 Smoothie	5	4	20	Credit Card		
7	3	3	9		Takeaway	
8 Sandwich	4	4	16	Cash		
9	5	3	15		Takeaway	
10 Sandwich	5	4	20		In-store	
11 Salad	5	5	25	Cash	Takeaway	
12 Sandwich	2	4	8	Credit Card	Takeaway	
13 Sandwich	2	4	8	Cash	In-store	
14 Cookie	5	1	5		Takeaway	
15	2	1.5	3		In-store	
16 Salad	3	5	15	Credit Card	In-store	
17 Sandwich	1	4	4			
18 Juice	2	3	6	Cash	In-store	
19 Cake	5	3	15	Cash		
20 Juice	4	3	12	Cash	Takeaway	
21 Smoothie		null	4	20 Cash	In-store	

Step 7: Recalculating the Total Spent Column

Power Query Applied Step: *Fix Total Spent*

Total Spent is mathematically dependent on Quantity and Price Per Unit, since Total Spent equals Quantity multiplied by Price Per Unit. Rows where Total Spent was missing or did not match this formula were recalculated using a custom column, ensuring the financial figures in the dataset are internally consistent.

Step 8: Recalculating the Quantity Column

Power Query Applied Step: *Fix Quantity*

Where Quantity was missing but both Total Spent and Price Per Unit were available, Quantity was derived as Total Spent divided by Price Per Unit and rounded to the nearest

Step 9: Recalculating the Price Per Unit Column

Power Query Applied Step: *Fix Price*

Similarly, where Price Per Unit was missing but Total Spent and Quantity were known, Price Per Unit was derived as Total Spent divided by Quantity. This three-way cross-checking across the previous three steps maximized the number of usable rows while keeping the numeric relationship between the three columns valid.

ABC	Total Spent Fixed	ABC	Quantity Fixed	ABC	Price Fixed
123	99% Valid 0% Error < 1% Empty	123	99% Valid 0% Error < 1% Empty	123	99% Valid 0% Error < 1% Empty
	4		2		2
	12		4		3
	4		4		1
	10		2		5
	4		2		2
	20		5		4
	9		3		3
	16		4		4
	15		5		3
	20		5		4
	25		5		5
	8		2		4
	8		2		4
	5		5		1
	15		3		5
	4		1		4
	6		2		3
	15		5		3
	12		4		3
	20		5		4

Step 10: Removing the Uncorrected Columns

Power Query Applied Step: *Remove Originals*

Once the corrected Quantity, Price Per Unit and Total Spent values had been calculated in new custom columns, the original, uncorrected versions of these columns were removed to avoid duplication and confusion in the final table.

Table.RemoveColumns(#Fix Price,{"Quantity", "Price Per Unit", "Total Spent"})

Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent Fixed	Quantity Fix
1	TXN_1961373	Coffee	Credit Card	Takeaway	08-09-2023	4
2	TXN_4977031	Cake	Cash	In-store	16-05-2023	12
3	TXN_4271903	Cookie	Credit Card	In-store	19-07-2023	4
4	TXN_7034554	Salad			27-04-2023	10
5	TXN_3160411	Coffee	Digital Wallet	In-store	11-06-2023	4
6	TXN_2602893	Smoothie	Credit Card		31-03-2023	20
7	TXN_4433211			Takeaway	06-10-2023	9
8	TXN_6699534	Sandwich	Cash		28-10-2023	16
9	TXN_4717867			Takeaway	28-07-2023	15
10	TXN_2064365	Sandwich	In-store		31-12-2023	20
11	TXN_2548360	Salad	Cash	Takeaway	07-11-2023	25
12	TXN_3051279	Sandwich	Credit Card	Takeaway	null	8
13	TXN_7619095	Sandwich	Cash	In-store	09-05-2023	8
14	TXN_3437049	Cookie		Takeaway	01-06-2023	5
15	TXN_8915701		In-store		21-03-2023	3
16	TXN_2847255	Salad	Credit Card	In-store	15-11-2023	15
17	TXN_3765707	Sandwich			10-06-2023	4
18	TXN_3769710	Juice	Cash	In-store	24-02-2023	6
19	TXN_3876618	Cake	Cash		25-03-2023	15
20	TXN_3709394	Juice	Cash	Takeaway	15-01-2023	12

Step 11: Finalizing Column Names

Power Query Applied Step: *Rename Columns*

The newly created corrected columns were renamed back to their original, business-friendly names, namely Quantity, Price Per Unit and Total Spent, so that the final table structure matches the expected schema, with clean values now sitting under the correct headers.

Total Spent	Quantity	Price Per Unit
4	2	2
12	4	3
4	4	1
10	2	5
4	2	2
20	5	4
9	3	3
16	4	4
15	5	3
20	5	4
25	5	5
8	2	4
8	2	4
5	5	1
3	2	1.5
15	3	5
4	1	4
6	2	3
15	5	3
12	4	3

Step 12: Reapplying Data Types

Power Query Applied Step: *Final Data Types*

After all value corrections and column renaming, data types were reapplied to guarantee that every column is stored in its correct, final format, whether Whole Number, Decimal Number, Date or Text, before the row-level cleaning steps that follow.

Payment Method	Location	Transaction Date	Total Spent	Quantity	Price Per Unit
Valid 68% Error 0% Empty 32%	Valid 60% Error 0% Empty 40%	Valid 95% Error 0% Empty 5%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%
4 distinct, 0 unique	3 distinct, 0 unique	366 distinct, 0 unique	18 distinct, 0 unique	6 distinct, 0 unique	7 distinct, 0 unique
Credit Card	Takeaway	08-09-2023	4	2	2
Cash	In-store	16-05-2023	12	4	3
Credit Card	In-store	19-07-2023	4	4	1
Digital Wallet	In-store	27-04-2023	10	2	5
Credit Card	In-store	11-06-2023	4	2	2
Cash	Takeaway	31-03-2023	20	5	4
Cash	Takeaway	06-10-2023	9	3	3
	Takeaway	28-10-2023	16	4	4
	Takeaway	28-07-2023	15	5	3
Cash	In-store	31-12-2023	20	5	4
Cash	Takeaway	07-11-2023	25	5	5
Credit Card	Takeaway	null	8	2	4
Cash	In-store	03-05-2023	8	2	4
	Takeaway	01-06-2023	5	5	1
	In-store	21-03-2023	3	2	1.5
Credit Card	In-store	15-11-2023	15	3	5
		10-06-2023	4	1	4
Cash	In-store	24-02-2023	6	2	3
Cash		25-03-2023	15	5	3
Cash	Takeaway	15-01-2023	12	4	3
Cash	In-store	04-04-2023	20	5	4

Step 13: Removing Invalid Dates

Power Query Applied Step: Filter Dates

Rows containing transaction dates that were blank, malformed, or fell outside the valid reporting period were filtered out. This ensures that any time-based analysis, such as monthly or yearly sales trends, is based only on valid, real transaction dates.

Transaction Date	Total Spent	Quantity	Price Per Unit
Valid 95% Error 0% Empty 5%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%
366 distinct, 0 unique	18 distinct, 0 unique	6 distinct, 0 unique	7 distinct, 0 unique
08-09-2023	4	2	
16-05-2023	12	4	
19-07-2023	4	4	
27-04-2023	10	2	
11-06-2023	4	2	
31-03-2023	20	5	
06-10-2023	9	3	
28-10-2023	16	4	
28-07-2023	15	5	
31-12-2023	20	5	
07-11-2023	25	5	
null	8	2	
03-05-2023	8	2	
01-06-2023	5	5	
21-03-2023	3	2	
15-11-2023	15	3	
10-06-2023	4	1	
24-02-2023	6	2	
25-03-2023	15	5	
15-01-2023	12	4	
04-04-2023	20	5	

Step 14: Handling Remaining Missing Item Values

Power Query Applied Step: Fill Item

For the Item column, remaining nulls left over after the UNKNOWN replacement step were handled using Power Query's Fill Down functionality, so that no product field was left blank in the final dataset.

Step 15: Handling Remaining Missing Payment Method Values

Power Query Applied Step: *Fill Payment*

The same fill logic was applied to the Payment Method column, ensuring that every transaction has a recorded payment method rather than a blank or placeholder value.

Step 16: Handling Remaining Missing Location Values

Power Query Applied Step: *Fill Location*

The Location column was treated in the same way, so that every transaction is clearly marked as either In-store or Takeaway with no blanks remaining.

Item	Payment Method	Location	Transaction Date	Total Spent	Quantity	Price
Coffee	Credit Card	Takeaway	08-09-2023	4	2	
Cake	Cash	In-store	26-05-2023	12	4	
Cookie	Credit Card	In-store	29-07-2023	4	4	
Salad	Unknown	Unknown	27-04-2023	10	2	
Coffee	Digital Wallet	In-store	21-06-2023	4	2	
Smoothie	Credit Card	Unknown	21-03-2023	20	5	
unknown Item	Unknown	Takeaway	06-10-2023	9	3	
Sandwich	Cash	Unknown	28-10-2023	16	4	
unknown Item	Unknown	Takeaway	28-07-2023	15	5	
Sandwich	Unknown	In-store	21-12-2023	20	5	
Salad	Cash	Takeaway	07-11-2023	25	5	
Sandwich	Credit Card	Takeaway	null	8	2	
Sandwich	Cash	In-store	09-05-2023	8	2	
Cookie	Unknown	Takeaway	01-06-2023	5	5	
unknown Item	Unknown	In-store	21-03-2023	3	2	
Salad	Credit Card	In-store	15-11-2023	15	3	
Sandwich	Unknown	Unknown	10-06-2023	4	1	
Juice	Cash	In-store	24-02-2023	6	2	
Cake	Cash	Unknown	25-03-2023	15	5	
Juice	Cash	Takeaway	15-01-2023	12	4	
Smoothie	Cash	In-store	04-04-2023	20	5	

Step 17: Removing Unrecoverable Records

Power Query Applied Step: *Remove Invalid Rows*

A small number of rows had multiple critical fields missing at once, for example both Quantity and Price Per Unit missing with no Total Spent to cross-check against, making them impossible to correct reliably. These rows were removed rather than filled with assumed values, to preserve the integrity of the analysis.

Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent	Quantity	Price
TXN_1961373	Coffee	Credit Card	Takeaway	08-09-2023	4	2	
TXN_4977011	Cake	Cash	In-store	26-05-2023	12	4	
TXN_4273903	Cookie	Credit Card	In-store	29-07-2023	4	4	
TXN_7036554	Salad	Unknown	Unknown	27-04-2023	10	2	
TXN_13160411	Coffee	Digital Wallet	In-store	21-06-2023	4	2	
TXN_2602893	Smoothie	Credit Card	Unknown	21-03-2023	20	5	
TXN_4433211	unknown Item	Unknown	Takeaway	06-10-2023	9	3	
TXN_6699534	Sandwich	Cash	Unknown	28-10-2023	16	4	
TXN_4717867	unknown Item	Unknown	Takeaway	28-07-2023	15	5	
TXN_2064365	Sandwich	Unknown	In-store	21-12-2023	20	5	
TXN_2548360	Salad	Cash	Takeaway	07-11-2023	25	5	
TXN_7619095	Sandwich	Cash	In-store	09-05-2023	8	2	
TXN_9437049	Cookie	Unknown	Takeaway	01-06-2023	5	5	
TXN_8915701	unknown Item	Unknown	In-store	21-03-2023	3	2	
TXN_2847235	Salad	Credit Card	In-store	15-11-2023	15	3	
TXN_3765707	Sandwich	Unknown	Unknown	10-06-2023	4	1	
TXN_6769710	Juice	Cash	In-store	24-02-2023	6	2	
TXN_8876618	Cake	Cash	Unknown	25-03-2023	15	5	
TXN_1705294	Juice	Cash	Takeaway	15-01-2023	12	4	
TXN_3522028	Smoothie	Cash	In-store	04-04-2023	20	5	
TXN_3567645	Smoothie	Credit Card	Takeaway	30-03-2023	16	4	

Step 18: Eliminating Duplicate Records

Power Query Applied Step: *Remove Duplicates*

The “Remove Duplicates” transformation was applied across the full row context to eliminate exact duplicate transaction records, ensuring that no sale was counted more than once in subsequent analysis.

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Transaction ID, Item, Payment Method, Location, Transaction Date, Total Spent, and Quantity. The table contains 21 rows of data. The 'APPLIED STEPS' pane on the right shows the 'Remove Duplicates' step selected, with the 'Source' property set to 'dirty_cafe_sales'. The 'PROPERTIES' pane on the right shows the 'Name' property set to 'dirty_cafe_sales'.

Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent	Quantity
TXN_1961373	Coffee	Credit Card	Takeaway	08-09-2023	4	4
TXN_4977031	Cake	Cash	In-store	16-05-2023	12	12
TXN_4271903	Cookie	Credit Card	In-store	19-07-2023	4	4
TXN_7034554	Salad	Unknown	Unknown	27-04-2023	10	10
TXN_3160411	Coffee	Digital Wallet	In-store	11-06-2023	4	4
TXN_2602893	Smoothie	Credit Card	Unknown	31-03-2023	20	20
TXN_4433211	unknown item	Unknown	Takeaway	06-10-2023	9	9
TXN_6695534	Sandwich	Cash	Unknown	28-10-2023	16	16
TXN_4712867	unknown item	Unknown	Takeaway	28-07-2023	15	15
TXN_2064365	Sandwich	Unknown	In-store	31-12-2023	20	20
TXN_2548360	Salad	Cash	Takeaway	07-11-2023	25	25
TXN_7618095	Sandwich	Cash	In-store	03-05-2023	8	8
TXN_3437049	Cookie	Unknown	Takeaway	01-06-2023	5	5
TXN_8915701	unknown item	Unknown	In-store	21-03-2023	3	3
TXN_2847255	Salad	Credit Card	In-store	15-11-2023	15	15
TXN_3765707	Sandwich	Unknown	Unknown	10-06-2023	4	4
TXN_6769710	Juice	Cash	In-store	24-02-2023	6	6
TXN_8876618	Cake	Cash	Unknown	25-03-2023	15	15
TXN_3709394	Juice	Cash	Takeaway	15-01-2023	12	12
TXN_3522028	Smoothie	Cash	In-store	04-04-2023	20	20
TXN_3567645	Smoothie	Credit Card	Takeaway	30-03-2023	16	16

Step 19: Removing Extra Spaces

Power Query Applied Step: *Trim Text*

The “Trim” transformation was applied to all text columns to remove leading, trailing, and repeated internal spaces, which otherwise cause identical-looking values, such as “Cash” and “ Cash,” to be treated as distinct categories.

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Transaction ID, Item, Payment Method, Location, Transaction Date, Total Spent, and Quantity. The table contains 21 rows of data. The 'APPLIED STEPS' pane on the right shows the 'Trim Text' step selected, with the 'Trim Text' property set to 'Trim Text'. The 'PROPERTIES' pane on the right shows the 'Name' property set to 'dirty_cafe_sales'.

Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent	Quantity
TXN_1961373	Coffee	Credit Card	Takeaway	08-09-2023	4	4
TXN_4977031	Cake	Cash	In-store	16-05-2023	12	12
TXN_4271903	Cookie	Credit Card	In-store	19-07-2023	4	4
TXN_7034554	Salad	Unknown	Unknown	27-04-2023	10	10
TXN_3160411	Coffee	Digital Wallet	In-store	11-06-2023	4	4
TXN_2602893	Smoothie	Credit Card	Unknown	31-03-2023	20	20
TXN_4433211	unknown item	Unknown	Takeaway	06-10-2023	9	9
TXN_6695534	Sandwich	Cash	Unknown	28-10-2023	16	16
TXN_4712867	unknown item	Unknown	Takeaway	28-07-2023	15	15
TXN_2064365	Sandwich	Unknown	In-store	31-12-2023	20	20
TXN_2548360	Salad	Cash	Takeaway	07-11-2023	25	25
TXN_7618095	Sandwich	Cash	In-store	03-05-2023	8	8
TXN_3437049	Cookie	Unknown	Takeaway	01-06-2023	5	5
TXN_8915701	unknown item	Unknown	In-store	21-03-2023	3	3
TXN_2847255	Salad	Credit Card	In-store	15-11-2023	15	15
TXN_3765707	Sandwich	Unknown	Unknown	10-06-2023	4	4
TXN_6769710	Juice	Cash	In-store	24-02-2023	6	6
TXN_8876618	Cake	Cash	Unknown	25-03-2023	15	15
TXN_3709394	Juice	Cash	Takeaway	15-01-2023	12	12
TXN_3522028	Smoothie	Cash	In-store	04-04-2023	20	20
TXN_3567645	Smoothie	Credit Card	Takeaway	30-03-2023	16	16

Step 20: Removing Non-Printable Characters

Power Query Applied Step: *Clean Text*

The “Clean” transformation was applied to strip any non-printable characters that may have been introduced during data export or import, further standardizing the text columns.

Transaction ID	Item	Payment Method	Location	Transaction Date	Total Spent	Quantity
9327 distinct, 9327 unique	9 distinct, 0 unique	4 distinct, 0 unique	3 distinct, 0 unique	345 distinct, 0 unique	18 distinct, 0 unique	6 distinct, 0 unique
TXN_1961373	Coffee	Credit Card	Takeaway	08-09-2023		4
TXN_4977031	Cake	Cash	In-store	16-05-2023		12
TXN_4271303	Cookie	Credit Card	In-store	19-07-2023		4
TXN_7034554	Salad	Unknown	Unknown	27-04-2023		10
TXN_3160411	Coffee	Digital Wallet	In-store	11-06-2023		4
TXN_2602893	Smoothie	Credit Card	Unknown	31-03-2023		20
TXN_4433211	unknown item	Unknown	Takeaway	06-10-2023		9
TXN_6695534	Sandwich	Cash	Unknown	28-10-2023		16
TXN_4737867	unknown item	Unknown	Takeaway	28-07-2023		15
TXN_2064365	Sandwich	Unknown	In-store	31-12-2023		20
TXN_2548360	Salad	Cash	Takeaway	07-11-2023		25
TXN_7613095	Sandwich	Cash	In-store	03-05-2023		8
TXN_9437049	Cookie	Unknown	Takeaway	01-06-2023		5
TXN_8915701	unknown item	Unknown	In-store	21-03-2023		3
TXN_2847255	Salad	Credit Card	In-store	15-11-2023		15
TXN_3765707	Sandwich	Unknown	Unknown	10-06-2023		4
TXN_6789710	Juice	Cash	In-store	24-02-2023		6
TXN_8876618	Cake	Cash	Unknown	25-03-2023		15
TXN_3709394	Juice	Cash	Takeaway	15-01-2023		12
TXN_3522028	Smoothie	Cash	In-store	04-04-2023		20
TXN_3567645	Smoothie	Credit Card	Takeaway	30-03-2023		16

Step 21: Removing Residual Blank Rows

Power Query Applied Step: *Remove Blanks*

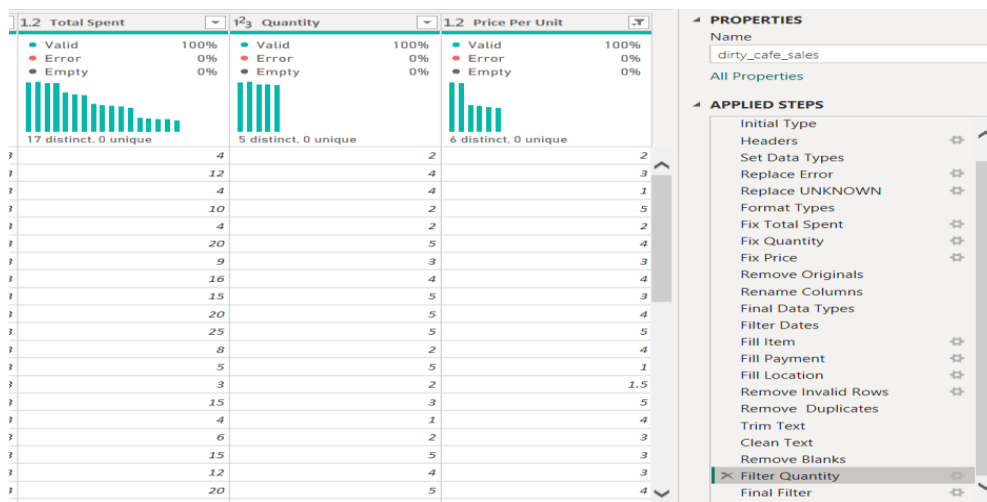
Any rows that were entirely blank across all columns, a common artifact of exporting data from spreadsheets, were removed from the table.

Payment Method	Location	Transaction Date	Total Spent	Quantity	Price Per Unit
4 distinct, 0 unique	3 distinct, 0 unique	345 distinct, 0 unique	17 distinct, 0 unique	5 distinct, 0 unique	6 distinct, 0 unique
Credit Card	Takeaway	08-09-2023		4	2
Cash	In-store	16-05-2023		12	4
Credit Card	In-store	19-07-2023		4	4
Unknown	Unknown	27-04-2023		10	2
Digital Wallet	In-store	11-06-2023		4	2
Credit Card	Unknown	31-03-2023		20	5
Unknown	Takeaway	06-10-2023		9	3
Cash	Unknown	28-10-2023		16	4
Unknown	Takeaway	28-07-2023		15	5
Unknown	In-store	31-12-2023		20	5
Cash	Takeaway	07-11-2023		25	5
Cash	In-store	03-05-2023		8	2
Unknown	Takeaway	01-06-2023		5	5
Unknown	In-store	21-03-2023		3	2
Credit Card	In-store	15-11-2023		15	3
Unknown	Unknown	10-06-2023		4	2
Cash	In-store	24-02-2023		6	2
Cash	Unknown	25-03-2023		15	5
Cash	Takeaway	15-01-2023		12	4
Cash	In-store	04-04-2023		20	5

Step 22: Removing Invalid Quantities

Power Query Applied Step: *Filter Quantity*

Rows with a Quantity of zero or a negative value were filtered out, as these do not represent valid, real-world sales transactions.



Step 23: Final Consistency Check

Power Query Applied Step: Final Filter

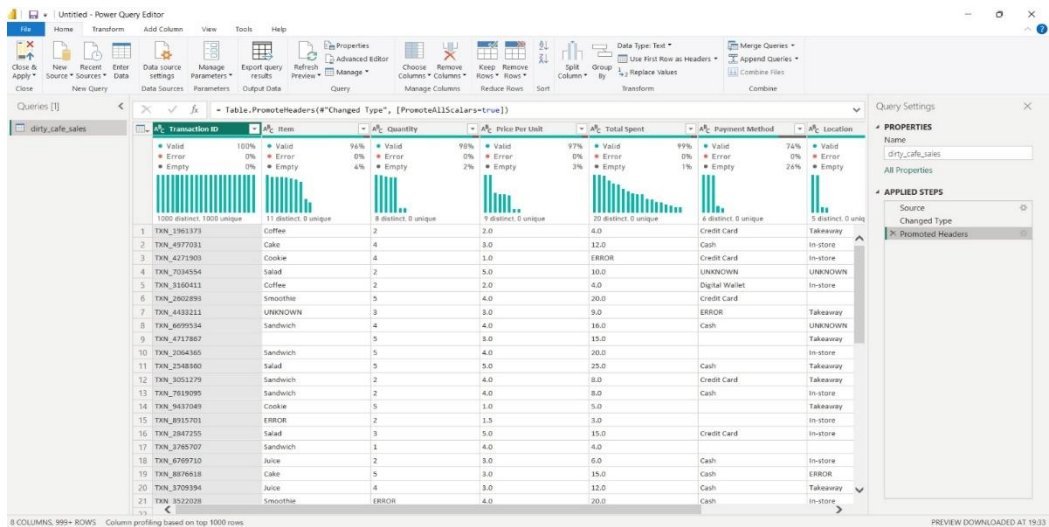
As a last safeguard, a final filter was applied across the cleaned table to catch any residual inconsistency, such as a Total Spent value of zero or a null that survived earlier steps, before the query results were loaded into the data model for analysis.



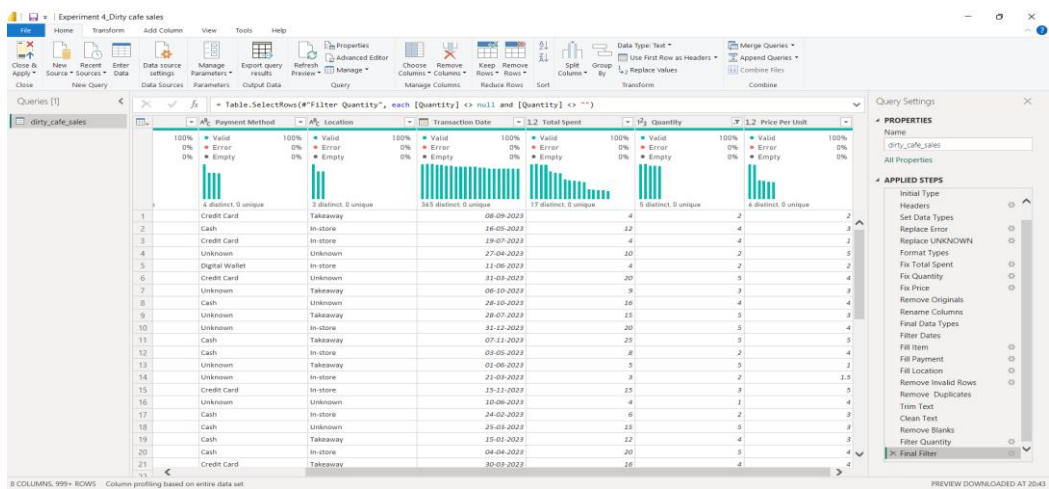
5. Before and After Comparison

The screenshots referenced below compare the state of the dataset before cleaning, at the raw import stage, and after cleaning, at the final output stage, illustrating the overall improvement in data quality.

5.1 Before Cleaning (Raw Data)

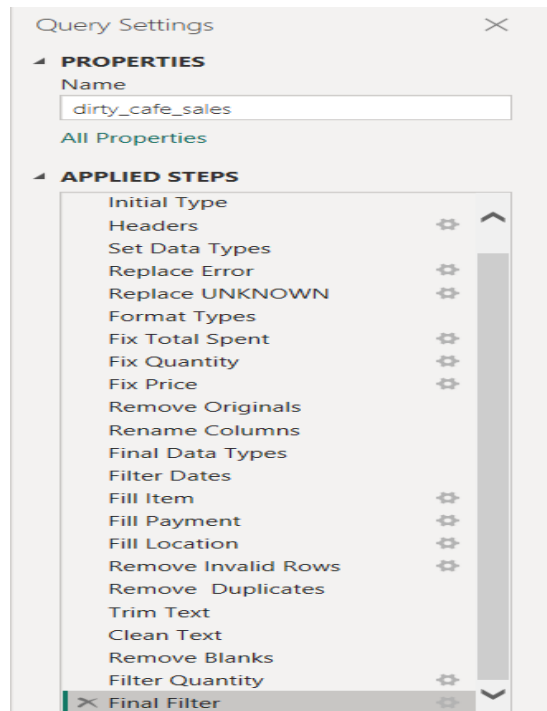


5.2 After Cleaning (Final Output)



5.3 Full List of Applied Steps

The screenshot below shows the complete Applied Steps pane from Power Query, confirming that every transformation described in this report was actually executed on the dataset.



6. Data Quality Summary

Before cleaning, the dataset contained several data quality issues, including **missing values**, **“UNKNOWN” placeholders**, **inconsistent data types**, **incorrect Total Spent calculations**, **duplicate records**, **extra spaces**, and **invalid dates and quantities**. These issues made the raw data inconsistent and unsuitable for reliable analysis.

After cleaning, missing and invalid values were appropriately handled, **“UNKNOWN” entries were replaced**, **data types were corrected**, and **Total Spent, Quantity, and Price Per Unit were made consistent**. Duplicate records were removed, text values were trimmed and cleaned, and invalid dates and quantities were filtered out. The resulting dataset is **clean, accurate, consistent, and ready for further analysis and visualization**.

7. Conclusion

Through a structured sequence of **twenty-three Applied Steps** in Power Query, the raw Cafe Sales dataset was successfully transformed into a **clean, consistent, and analysis-ready dataset**. The cleaning process addressed important issues such as incorrect data types, errors, missing values, placeholder entries, incorrect calculations, duplicate records, and invalid rows.

Since each transformation is saved as a **named and reusable Applied Step**, the same cleaning logic can be reapplied whenever the source data is refreshed or updated. The final dataset can therefore be used confidently for further analysis, including **sales trend analysis, payment method distribution, and location-based performance comparison**.

8. Key Learnings

- **Real-world datasets are rarely clean by default**, so proper data preparation is an important part of any analysis.

- **Missing values and placeholders** such as “UNKNOWN” and “ERROR” should be identified and handled carefully to maintain data consistency.
- Existing relationships between columns can help recover incorrect or missing values. For example, **Total Spent = Quantity × Price Per Unit** can be used to verify and correct values.
- Dividing the cleaning process into **small, clearly named steps** makes it easier to understand, review, and troubleshoot.
- Keeping a record of each transformation makes the workflow **transparent, repeatable, and auditable**.
- The exercise also showed the importance of explaining **why a particular cleaning step is performed**, rather than only documenting the actions taken.
- A well-cleaned dataset provides a **reliable foundation for accurate analysis and meaningful business insights**.